BrandGuard AI

Round 2 Submission: The Build

Team: ICYBITS

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Solo submission

Live deployment: https://brand-gov.vercel.app

1. Working Core Workflow

The end-to-end journey works from start to finish:

- Register / login with a role (creator, reviewer, manager, steward, admin).
- Brand Steward defines the brand kit: guidelines, approved facts, terminology, assets.
- Creator creates a draft (text paste or file upload: PDF, DOCX, image).
- Creator submits for review; the AI reviews against the brand kit and produces findings.
- Reviewer/manager triages findings: accept, edit or dismiss (reason required).
- Creator revises; revision history recorded.
- Manager approves; content is published.
- Insights dashboard aggregates recurring issues and unclear guidelines.

2. Completed Features

Area	Feature	Status
Auth	Register, login, logout, role-based sessions	Done
Brand Kit	Guidelines, approved facts, terminology, assets (CRUD)	Done
Content	Create draft, text + file upload (PDF/DOCX/image), edit body	Done
AI Review	LLM review against brand kit, structured findings	Done
Issues	Category, severity, location, explanation, source, suggested fix, confidence	Done
Workflow	Submit, review decision, comments, revision history, publish	Done
Insights	Counts by category/severity, content/published totals	Done
Confidentiality	Role-based access, org isolation, AI disclosure banner	Done

3. Technology and Implementation

Stack: Next.js 16 (App Router, TypeScript, Tailwind CSS), Prisma 5 + PostgreSQL (production) / SQLite (local dev), Groq LLM (OpenAI-compatible API, gpt-oss-20b), pdf-parse, mammoth, image-size, bcryptjs, jose (JWT sessions), Recharts.

Data flow: content submission -> file parsing -> LLM review with brand-kit context -> validated JSON findings stored as Issue records -> human review decisions -> revision tracking -> publish. All AI output is advisory; nothing is auto-applied.

AI components: structured prompt with brand kit slice; output constrained to a finding JSON schema; fenced-markdown stripping and array extraction fallback; max_tokens 4000; temperature 0.2.

4. Changes from Round 1 Plan

- Switched from the planned FastAPI/Python parsing to Next.js API routes with pdf-parse/mammoth for a single deployable unit.
- SQLite for local dev; deployed on PostgreSQL (Neon) in production on Vercel.
- Multi-tenant hardening shipped in Round 2: role guards (403), org checks on every single-item route, forced-creator registration, issue-status whitelist, re-review sync preserving decisions, creator-only body edits.
- Added a review-screen mockup and finding schema (originally planned for Round 3) to strengthen Round 2.
- Insights view is a first version: counts by category/severity; trend charts and heatmap moved to Round 3.

5. Three Major Improvements for Round 3

#	Improvement	Why it matters
1	Trend analytics + recurring-issue detection with charts and an 'unclear guideline' flag	Turns review data into brand-kit improvements over time.
2	Richer visual review: image alt-text/contrast checks, asset-match suggestions, dimension/format validation	Closes the visual-consistency and accessibility gaps.
3	Audit export + on-premise option (multi-tenant hardening delivered in Round 2: role guards, org isolation, RBAC)	Protects confidential materials at scale; hardening already shipped.

6. Usability Notes

Demo accounts (password: demo123): admin@demo.com, steward@demo.com, creator@demo.com, reviewer@demo.com, manager@demo.com. A brand kit is pre-seeded. Live app: https://brand-gov.vercel.app. AI disclosure banner is shown in the nav. Findings are cards with severity, source reference, suggested fix and accept/edit/dismiss actions.

7. How to Run (for judges)

```
npm install
npx prisma db push
node prisma/seed.js
npm run dev
# open http://localhost:3000
# or use the live deployment: https://brand-gov.vercel.app
# demo accounts (password: demo123): admin@, steward@, creator@, reviewer@, manager@demo.com
```

8. Limitations (honest)

- AI accuracy depends on brand-kit quality; small golden set not yet built.
- Visual checks are heuristic (alt text, contrast hints), not full design review.
- No email notifications; in-app only.
- Multi-tenant isolation is logical (orgId) and enforced at the route level; full audit export remains for Round 3.